Intensity, decibels and Doppler

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish intensity change from frequency shift.

SUCCESS CRITERIA - I can...

1. Use an intensity ratio separately from pitch.
2. Interpret a logarithmic sound-level difference.
3. Predict approach and recession Doppler trends.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **500 Hz** sound travels through air. Do air particles mainly move parallel or perpendicular to propagation? **Name the wave type**.

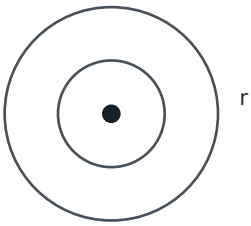
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta\beta = \mathbf{10}\ \log_{10}(I_2/I_1)$.
I is intensity; β is sound level in dB. Tenfold intensity gives **+10 dB**. Approach generally raises observed frequency; recession lowers it.

Worked reasoning: For the same point source, doubling distance spreads its power over four times the area, so ideal intensity drops to one quarter. This assumes spherical spreading without extra absorption.

Your turn: Draw two concentric wavefronts and indicate why the outer surface shares the power more widely.



Label distances and compare intensity.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A moving source approaches a stationary listener, then recedes. **Sketch closer wavefront spacing** in front and wider spacing behind; **compare heard pitch**.

Source moves right. Add wavefronts.



Marzano check - circle: ① ② ③

04 Apply the physics

A sound meter comparison uses the rule: multiplying intensity by **10** adds **10 dB**. **Complete the table** for intensity multipliers **10** and **100**. **Explain why 100** times intensity is not a **100 dB** increase.

Case	Intensity ratio	Level change (dB)
A	___	___
B	___	___

Marzano check - circle: ① ② ③

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05 Explain your reasoning

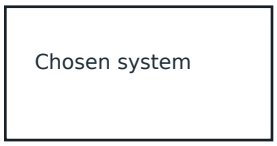
A loud stationary siren and a quiet approaching siren are compared. Does “louder” alone mean “higher frequency”? Explain **0**.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A sample warms by **4°C** after a certain energy input. If absorbed energy becomes **6** times as large with the same mass and material and no phase change, find the new rise.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

An observer hears a steadily emitting siren approach. Is observed frequency higher or lower than emitted frequency?

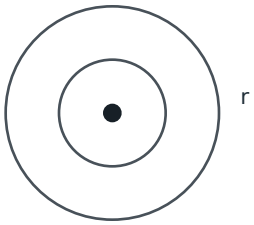
Source moves right. Add wavefronts.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Compare intensities at **1**, **2** and **3 m** from the same ideal source as fractions of the **1 m** value.



Label distances and compare intensity.

Marzano check - circle: 1 2 3